

# Why a $6 \times 10^{-3}$ RWA error moves the CNOT gate fidelity by only $2 \times 10^{-6}$

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## 1 The apparent paradox

The CNOT3 convergence experiment solves the same physical gate problem in the rotating frame with the rotating wave approximation (**rwa**) and without it (**norwa**, exactly equivalent to the laboratory frame). The convergence figures mark the “RWA modeling error,” the 2-norm difference between the two converged final states,

$$\|\delta\| := \|U_{\text{norwa}} - U_{\text{rwa}}\|_F = 6.008 \times 10^{-3}, \quad (1)$$

as the accuracy floor below which further numerical refinement of the RWA model is pointless. Yet the CNOT gate fidelities of the two solutions, measured against the same target, are nearly identical:

$$1 - F_{\text{rwa}} = 4.0269 \times 10^{-5}, \quad 1 - F_{\text{norwa}} = 4.2667 \times 10^{-5}, \quad (2)$$

a difference of only  $2.398 \times 10^{-6}$ , some 2500 times smaller than  $\|\delta\|$ . Both numbers are correct; they measure different things, and the gap between them is the product of two geometric effects. This note works them out and verifies each one quantitatively against the computed solutions.

**Setup.** The system has  $N = 160$  levels (a 10-level storage cavity and two 4-level transmons) and  $m = 4$  essential computational states. A solution is the matrix  $U \in \mathbb{C}^{N \times m}$  whose columns are the essential basis states propagated to the final time  $T = 550$  ns; unitarity gives  $\|U\|_F = \sqrt{m}$ . The target is the isometry  $V \in \mathbb{C}^{N \times m}$  (satisfying  $V^\dagger V = I_m$ ) whose columns are the CNOT images of the basis states, expressed in the appropriate frame. The gate fidelity is the standard trace overlap

$$F(U) = \frac{|z|^2}{m^2}, \quad z := \text{tr}(V^\dagger U) = \langle V, U \rangle_F. \quad (3)$$

Because the columns of  $V$  are supported only on the essential levels, the overlap automatically projects out guard-level leakage; leakage lowers  $|z|$  and hence  $F$ .

## 2 Effect 1: infidelity is quadratic in the miss distance

Fidelity is blind to a global phase, so the natural distance from a solution to the target is the distance to the whole phase orbit  $\{e^{i\varphi}V : \varphi \in [0, 2\pi)\}$ :

$$d(U) := \min_{\varphi} \|U - e^{i\varphi}V\|_F. \quad (4)$$

Expanding the square and minimizing (the optimum is  $e^{i\varphi^*} = z/|z|$ ) gives  $d^2 = \|U\|_F^2 + \|V\|_F^2 - 2|z| = 2m - 2|z|$ , i.e.  $|z| = m - d^2/2$ . Substituting into (3),

$$1 - F = \frac{d^2}{m} \left(1 - \frac{d^2}{4m}\right) \approx \frac{d^2}{m} \quad (d \ll 1). \quad (5)$$

Infidelity is *quadratic* in how far the solution sits from the target. A miss of size  $10^{-2}$  therefore lives on the  $10^{-5}$  infidelity scale, with no error mechanism required beyond the change of units. Numerically,

$$d_{\text{rwa}} = 1.2692 \times 10^{-2} \Rightarrow \frac{d_{\text{rwa}}^2}{4} = 4.0270 \times 10^{-5}, \quad (6)$$

which reproduces the measured  $1 - F_{\text{rwa}} = 4.0269 \times 10^{-5}$  to four digits (the small deficit is the  $d^4$  term). Likewise  $d_{\text{norwa}} = 1.3064 \times 10^{-2}$  gives  $4.267 \times 10^{-5}$ .

Effect 1 alone does not resolve the paradox: if the RWA perturbation  $\delta$  pushed the solution straight away from the target, the infidelity would change by up to  $(2d\|\delta\| + \|\delta\|^2)/m = 4.7 \times 10^{-5}$ , i.e. it would roughly double. The observed change is twenty times smaller than that bound, which is the second effect.

### 3 Effect 2: fidelity sees only the radial component

The infidelity (5) depends on the state only through the scalar  $d$ , the distance to the target orbit. Under a perturbation  $U \mapsto U + \delta$ , the overlap shifts by  $\langle V, \delta \rangle$  and, to first order,

$$\Delta|z| = \text{Re} \left[ \frac{\bar{z}}{|z|} \langle V, \delta \rangle \right] + O(\|\delta\|^2), \quad (7)$$

the projection of  $\delta$  onto a *single real direction* out of the  $2Nm = 1280$  real dimensions of the state space: the “radial” direction  $\hat{r} = (U - e^{i\varphi^*}V)/d$  pointing from the phase-aligned target to the solution. Every other direction is tangential and invisible at first order. (Since  $d$  is a distance function, it is also globally 1-Lipschitz,  $|\Delta d| \leq \|\delta\|$ , so radial alignment is the worst case, not just the first-order one.)

Decomposing the measured RWA perturbation  $\delta$  at  $U_{\text{rwa}}$  into mutually orthogonal pieces gives Table 1.

| component        | direction              | size                   | share of $\ \delta\ ^2$ |
|------------------|------------------------|------------------------|-------------------------|
| global phase     | $iU/\ U\ _F$           | $5.211 \times 10^{-3}$ | 75.2%                   |
| other tangential | (orthogonal remainder) | $2.990 \times 10^{-3}$ | 24.8%                   |
| radial           | $\hat{r}$              | $2.58 \times 10^{-5}$  | 0.002%                  |
| total            |                        | $6.008 \times 10^{-3}$ | 100%                    |

Table 1: Orthogonal decomposition of the RWA perturbation  $\delta = U_{\text{norwa}} - U_{\text{rwa}}$ .

Three-quarters of  $\|\delta\|^2$  is a *relative global phase*: the two solutions differ by  $e^{i\Delta\varphi}$  with  $\Delta\varphi = 2.6$  mrad, equivalent to a common level shift of under a kilohertz integrated over the 550 ns gate. Fidelity is exactly invariant under this component, at all orders. Almost all of the rest is tangential: it changes the *direction* in which the solution misses the target, not the distance. A

tangential step of length  $\tau$  does grow the distance at second order, through the curvature of the sphere of radius  $d$  around the target orbit,  $\Delta d \approx \tau^2/(2d)$ , and this closes the books exactly:

$$\underbrace{\frac{(2.990 \times 10^{-3})^2}{2(1.2692 \times 10^{-2})}}_{\text{tangential curvature}} + \underbrace{2.58 \times 10^{-5}}_{\text{radial}} = 3.78 \times 10^{-4} \approx \Delta d_{\text{measured}} = 3.72 \times 10^{-4}, \quad (8)$$

and then, by (5),

$$\Delta(1 - F) = \frac{d_{\text{norwa}}^2 - d_{\text{rwa}}^2}{m} = 2.398 \times 10^{-6}, \quad (9)$$

which is the measured infidelity difference.

The tiny radial component is not a coincidence of tuning. A perturbation pointing in a *random* direction of the 1280-dimensional real state space would have an expected radial fraction of order  $1/\sqrt{1280} \approx 2.8\%$ ; the measured first-order radial fraction is 0.43%. The counter-rotating terms have no knowledge of the particular direction in which the optimized pulse happens to miss the target, so their effect is generically tangential. Physically this is the expected structure of the RWA error: the leading effect of the counter-rotating terms is a small coherent energy shift (of Bloch–Siegert type [5, 6]), whose dominant consequence over a long gate is accumulated phase. The common part of that phase is pure gauge (the 75% row of Table 1), and the differential part redistributes the miss rather than enlarging it.

## 4 Reconciling the two plots

Both numbers are the right answer to different questions. The 2-norm line  $\|\delta\| = 6 \times 10^{-3}$  answers: *below what numerical error does refining the RWA model stop improving agreement with the true (lab-frame) state?* For full-state observables, phases included, the RWA really is a  $6 \times 10^{-3}$ -level modeling error. The fidelity comparison answers: *does the RWA change how good the gate is?* There the two effects stack: the quadratic map (5) converts  $10^{-2}$ -scale distances to  $10^{-5}$ -scale infidelities, and the radial projection suppresses the sensitivity to this particular perturbation by another factor of twenty, leaving  $2.4 \times 10^{-6}$ . A useful mnemonic for the overall bookkeeping:

$$\underbrace{6.0 \times 10^{-3}}_{\|\delta\|} \xrightarrow{\text{drop global phase}} 3.0 \times 10^{-3} \xrightarrow{\text{curvature } \tau^2/2d} \Delta d = 3.7 \times 10^{-4} \xrightarrow{2d \Delta d/m} 2.4 \times 10^{-6}. \quad (10)$$

## 5 Relation to the literature

The two ingredients are individually well documented, though we are not aware of a reference that presents this specific side-by-side observation (state-norm RWA error versus gate-fidelity RWA effect) for an optimized multilevel gate.

*Quadratic gap for coherent errors.* Effect 1 is the same mathematics behind a well-known result in the gate-benchmarking literature: for coherent (unitary) errors the worst-case error rate (diamond norm) scales as the *square root* of the average gate infidelity, so norm-type error measures and fidelity-type measures of the same coherent error can differ by orders of magnitude [1, 2, 3]. That literature runs the argument in the direction “reported fidelities

flatter coherent errors”; here the same fact appears in reverse, as a plotted norm error that looks alarming on a fidelity scale.

*Stationarity of fidelity.* Effect 2 is a consequence of  $F$  being stationary in every direction but one; this is the elementary geometry underlying the quantum control landscape literature, where the fidelity’s critical structure as a function of the evolution is analyzed in detail [4].

*Character of the RWA error.* That the leading counter-rotating effect is a coherent, largely calibratable phase shift goes back to Bloch and Siegert [5]; modern treatments include the exact rotating frame construction of Zeuch et al. [6] and recent gate-focused analyses that compute counter-rotating contributions to infidelity perturbatively and cancel them with pulse corrections or phase calibration [7, 8]. Those works are consistent with Table 1: the RWA error is dominated by phase accumulation, which gate fidelity largely forgives.

## References

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- [7] D. A. Rower et al., *Suppressing counter-rotating errors for fast single-qubit gates with fluxonium*, PRX Quantum **5**, 040342 (2024).
- [8] *Single-qubit gates beyond the rotating-wave approximation for strongly anharmonic low-frequency qubits*, arXiv:2503.08238 (2025).

## A Reproducing the numbers

All quantities in this note come from the cached Vern9 references (`abstol = reltol = 10-15`, basis initial condition,  $T = 550$ ,  $N_{\text{osc}} = 10$ , two guard levels per transmon) via `vern9_reference` and `cnot3_gate_fidelity` in `src/cnot3_reference.jl`. The decomposition of Table 1 projects  $\delta = U_{\text{norwa}} - U_{\text{rwa}}$  onto the gauge direction  $iU_{\text{rwa}}/\|U_{\text{rwa}}\|_F$  and the (gauge-orthogonalized) radial direction  $(U_{\text{rwa}} - e^{i\varphi^*}V)/d$ , using the real inner product  $\text{Re}\langle \cdot, \cdot \rangle_F$ ; the tangential remainder is obtained by Pythagoras.